



Modelling Financial Markets

Seminar 15: LLMs and Sentiment Analysis — Hands-on



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Acronyms used in this material (1/2)

- AI: Artificial Intelligence
- AUC: Area Under the (ROC) Curve
- BERT: Bidirectional Encoder Representations from Transformers
- BVB: Bursa de Valori București (the Bucharest Stock Exchange)
- CAR: Cumulative Abnormal Return
- CI: Confidence Interval
- DAX: Deutscher Aktienindex (the German stock index (Frankfurt Stock Exchange))
- DELTA: Deep Learning for Text Analytics (Quantinar course)
- ECE: Expected Calibration Error
- ES: Expected Shortfall
- FNSPID: Financial News and Stock Price Integration Dataset
- GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- GI: (Harvard) General Inquirer, dictionary IV-4
- GPT: Generative Pre-trained Transformer
- GPT-3: Generative Pre-trained Transformer 3 (here version 3.5)
- GPT-4: Generative Pre-trained Transformer 4
- IV-4: Harvard IV-4 (version 4 of the General Inquirer word lists)
- LDA: Latent Dirichlet Allocation
- LLM: Large Language Model
- LM: Loughran-McDonald (financial sentiment dictionary)
- NLL: Negative Log-Likelihood
- OLS: Ordinary Least Squares
- POF: Proportion Of Failures (test)
- PPI: Prediction-Powered Inference
- RLHF: Reinforcement Learning from Human Feedback
- ROC: Receiver Operating Characteristic (curve)

Acronyms used in this material (2/2)

- **SE: Standard Error**
- **TF-IDF: Term Frequency - Inverse Document Frequency**
- **US: United States**
- **UTC: Coordinated Universal Time (Temps Universel Coordonné)**

Outline

Introduction

Part A: Derivations with Numbers

Part B: Estimation and Inference

Part C: Open Question and AI Critique

Wrap-up

References

Seminar Objectives

By the end of this seminar you can

- ▣ Score a text with a sentiment dictionary and with the label probabilities of a large language model (LLM)
- ▣ Derive the attenuation of a regression slope on a misclassified sentiment label, and the variance and power of paired classifier tests
- ▣ Compare two classifiers with a paired test and detect training-data leakage and look-ahead bias
- ▣ Make inference valid with model scores: prediction-powered inference, wild cluster bootstrap with few clusters, multiplicity and specification curves

- ▣ Background: Lecture 15; textbook [Franke, Härdle & Hafner \(2019\)](#)

Structure

- ▣ **Part A:** eight derivations with numbers (A1–A8)
- ▣ **Part B:** eleven computational exercises with inference (B1–B11)
- ▣ **Part C:** two open questions, seeds for the team project (C1, and C3 from our research on LLM-VaR), and the critique of an AI answer (C2)

How to Work in This Seminar

- **[Solved]** problems show the full worked solution; **[Proposed]** problems show only the task: their solutions are discussed in class
- Every proposed problem names its **model**: the solved problem that uses the same technique
 - ▶ Every Part B exercise ends with interpretation questions
- You may use an AI assistant, but declare it in [AI_USE.md](#) and log the errors you catch in [AI_ERRORS.md](#)
 - ▶ C2 trains exactly this: find the errors in an AI answer about sentiment backtests
- Always ask: was the text public before the return I try to predict, and did the model already know the outcome?

Solved and Proposed Problems

Solved: worked examples in this deck

- Part A: A1 (attenuation from misclassification), A3 (variance and power of a paired test), A5 (probabilities and the optimal temperature), A7 (break-even cost with SE)
- Part B: B1 (FinBERT vs LM, McNemar), B3 (training-data leakage), B5 (clustered standard errors), B7 (does the LLM remember the market?), B9 (prediction-powered inference)

Proposed: you solve them

- Part A: A2 (LM attenuation; model: A1), A4 (power, FinBERT vs Qwen; model: A3), A6 (Fleiss' κ , Dawid–Skene; model: A1), A8 (Qwen after costs; model: A7)
- Part B: B2 (model: B1), B4 (model: B3), B6 (model: B5), B8 (model: B1), B10 (wild cluster bootstrap; model: B5), B11 (specification curve; model: B7)
- Part C: C1, does headline sentiment pay after costs?; C2, critique an AI answer on an LLM sentiment backtest; C3, LLM-VaR on course data (model: B7)

Setup: Python Environment and Data

-  **Open in Colab:** [chapter15_seminar_notebook.ipynb](#)
- Google Colab (recommended) or local Python ≥ 3.10 with `numpy`, `pandas`, `scipy`, `scikit-learn`, `matplotlib`; `transformers` only to recompute the scores
- Labelled texts: Financial PhraseBank (4 846 news sentences, 2 264 with full agreement) [Malo et al. \(2014\)](#); Twitter Financial News, validation set (2 388 headlines)
- Dated headlines: FNSPID (Financial News and Stock Price Integration Dataset) [Dong, Fan & Peng \(2024\)](#), 106 820 headlines on 16 large US stocks; daily closes of the stocks and of SPY (an S&P 500 index fund)
- The scores of every model are precomputed in the tables of Chapter 15; Quantlets: [MFM_ch15_sem_partA](#), [partB](#), [partC](#)

Formulas Used Today (1/2): Scores and Metrics

- Dictionary tone: $\tau = (P - N)/(P + N)$, P , N : counts of positive and negative words; $\tau = 0$ if $P + N = 0$
 - ▶ LM: the Loughran–McDonald financial word lists [Loughran & McDonald \(2011\)](#); GI: the Harvard General Inquirer IV-4 list, a generic psychology dictionary
- LLM label probabilities: $p_k = e^{z_k/T} / \sum_j e^{z_j/T}$ (z_k : logit of answer word k , T : temperature); net score $s = p_{\text{pos}} - p_{\text{neg}}$
 - ▶ FinBERT: a BERT (Bidirectional Encoder Representations from Transformers) model fine-tuned for financial sentiment [Araci \(2019\)](#); Qwen2.5: open-weight LLMs [Qwen Team \(2024\)](#)
- Misclassified binary regressor [Aigner \(1973\)](#): $\text{plim } \hat{b} = \beta \pi(1 - \pi)(1 - \alpha_0 - \alpha_1)/[p(1 - p)]$
 - ▶ Fleiss' $\kappa = (\bar{P} - \bar{P}_e)/(1 - \bar{P}_e)$ for agreement among many raters [Fleiss \(1971\)](#)

Formulas Used Today (2/2): Inference and Signals

- McNemar test [McNemar \(1947\)](#): n_{01} texts where A is wrong and B right, n_{10} the reverse; under H_0 , $n_{01} \sim \text{Bin}(n_{01} + n_{10}, 0.5)$
 - Bootstrap CI (confidence interval): resample texts 2000 times, take the 2.5% and 97.5% quantiles
- AUC (area under the ROC curve): probability that a random “up” case gets a higher score than a random “down” case; 0.5 = no information
- PPI [Angelopoulos et al. \(2023\)](#), n labelled and N unlabelled texts sampled independently, f fixed:

$$\hat{\theta} = \frac{1}{N} \sum_{\text{unlab.}} f(\tilde{X}_i) - \frac{1}{n} \sum_{\text{lab.}} (f(X_j) - Y_j), \text{ variance } \widehat{\text{Var}}(f)/N + \widehat{\text{Var}}(f - Y)/n$$
- Panel regression $r_{i,d+h}^e = a + b z_{i,d} + \varepsilon$: r^e excess return over SPY (%), z standardised daily score, $h = 0, 1, 2$ days after the news day d
 - Standard errors (SE): ordinary least squares (OLS) or clustered by day or by stock [Petersen \(2009\)](#); bp: basis point, 0.01%
- Strategy: Newey–West t [Newey & West \(1987\)](#); annualised Sharpe ratio $\sqrt{252} \bar{r}/s_r$; break-even cost = mean gross return on days with positions / 4 (four trades a day)

A1: Attenuation from Misclassification (FinBERT) (task 1/2) [Solved]

Task

□ Context and assumptions

- ▶ Data: Twitter Financial News, validation set of $N = 2\,388$ headlines published in 2022, each labelled negative, neutral or positive by annotators; the human label is treated as the truth
- ▶ Regression of interest: $r = a + \beta s^* + \varepsilon$, where $s^* = 1$ if a headline is truly negative and 0 otherwise; we only observe $\hat{s} = 1$ when FinBERT predicts “negative”
- ▶ Assumption (non-differential error): once s^* is known, $\hat{s}$ carries no information about r , so $\text{Cov}(\varepsilon, \hat{s}) = 0$
- ▶ Notation: $\pi = P(s^* = 1)$, $p = P(\hat{s} = 1)$, false-alarm rate $\alpha_0 = P(\hat{s} = 1 \mid s^* = 0)$, miss rate $\alpha_1 = P(\hat{s} = 0 \mid s^* = 1)$; FinBERT confusion matrix (rows: true label, columns: prediction):

True \ predicted	neg	neu	pos	Total
neg	264	71	12	347
neu	213	1 184	169	1 566
pos	58	133	284	475
Total	535	1 388	465	2 388

A1: Attenuation from Misclassification (FinBERT) (task 2/2) [Solved]

Task

What to do

- ▶ (a) Write the probability limit of the OLS slope of r on $\hat{s}$ and show that it equals $\beta \text{Cov}(s^*, \hat{s}) / \text{Var}(\hat{s})$ under the assumption above
- ▶ (b) Derive $\text{Cov}(s^*, \hat{s}) = \pi(1 - \pi)(1 - \alpha_0 - \alpha_1)$ and $\text{Var}(\hat{s}) = p(1 - p)$, hence the attenuation factor $\lambda = \pi(1 - \pi)(1 - \alpha_0 - \alpha_1) / [p(1 - p)]$
- ▶ (c) Compute π , p , α_0 and α_1 from the first row and the first column of the matrix, then λ to three decimal places
- ▶ (d) Compute $\text{Cov}(u, s^*)$ for the measurement error $u = \hat{s} - s^*$ and decide whether the error is classical (uncorrelated with s^*)

What to report

- ▶ λ and the factor $1/\lambda$ by which the true β exceeds the observed slope [Aigner \(1973\)](#)
- ▶ Interpretation: what does λ imply for a slope on the FinBERT negative-news dummy, and why does it not carry over to the continuous-score slopes of Lecture 15?

A1: Solution (1/2) [Solved]

Solution

- (a) For one regressor, $\text{plim } \hat{b} = \text{Cov}(r, \hat{s}) / \text{Var}(\hat{s})$; substituting $r = a + \beta s^* + \varepsilon$ gives $\text{Cov}(r, \hat{s}) = \beta \text{Cov}(s^*, \hat{s}) + \text{Cov}(\varepsilon, \hat{s})$
- The error is non-differential, so $\text{Cov}(\varepsilon, \hat{s}) = 0$ and $\text{plim } \hat{b} = \beta \text{Cov}(s^*, \hat{s}) / \text{Var}(\hat{s})$
- (b) Both variables are binary: $\text{Cov}(s^*, \hat{s}) = P(s^* = 1, \hat{s} = 1) - \pi p = \pi(1 - \alpha_1) - \pi p$
- By total probability $p = \pi(1 - \alpha_1) + (1 - \pi)\alpha_0$; substituting,
 $\text{Cov}(s^*, \hat{s}) = \pi(1 - \pi)(1 - \alpha_1) - \pi(1 - \pi)\alpha_0 = \pi(1 - \pi)(1 - \alpha_0 - \alpha_1)$
- $\hat{s}$ is a Bernoulli variable, so $\text{Var}(\hat{s}) = p(1 - p)$, and the ratio of the two gives λ

A1: Solution (2/2) [Solved]

Solution

- (c) $\pi = 347/2\,388 = 0.1453$ and $p = 535/2\,388 = 0.2240$
- $\alpha_1 = 1 - 264/347 = 0.2392$ (missed negatives); $\alpha_0 = 271/2\,041 = 0.1328$ (neutral or positive headlines called negative)
- $\lambda = \frac{0.1242 \times 0.6280}{0.2240 \times (1 - 0.2240)} = \frac{0.0780}{0.1738} = 0.449$, so $\beta = 2.23 \hat{b}$
- (d) $\text{Cov}(u, s^*) = \text{Cov}(\hat{s}, s^*) - \text{Var}(s^*) = 0.0780 - 0.1242 = -0.0462 < 0$: the error is negatively correlated with the truth, so it is not classical [Aigner \(1973\)](#)
- Interpretation: a slope on the FinBERT dummy keeps only about 45% of the true effect; the Lecture 15 slopes use a standardised continuous score, attenuated by $\rho(s^*, \hat{s})$ instead (lecture), so λ does not transfer

A2: Attenuation for the LM Dictionary (task 1/2) [Proposed]

Solution discussed in class

Task

□ Context and assumptions

- ▶ Data: Twitter Financial News, validation set of $N = 2\,388$ headlines published in 2022, each labelled negative, neutral or positive by annotators; the human label is treated as the truth
- ▶ Same regression and the same non-differential assumption as in A1; now $\hat{s} = 1$ when the Loughran–McDonald (LM) tone of the headline is negative
- ▶ LM confusion matrix on the same headlines (rows: true label, columns: prediction):

True \ predicted	neg	neu	pos	Total
neg	172	166	9	347
neu	247	1 192	127	1 566
pos	45	349	81	475
Total	464	1 707	217	2 388

A2: Attenuation for the LM Dictionary (task 2/2) [Proposed]

Solution discussed in class

Task

□ What to do

- ▶ (a) Compute π , p , α_0 and α_1 from the first row and the first column of the matrix
- ▶ (b) Compute the attenuation factor λ and the correction factor $1/\lambda$ with the formula of A1
- ▶ (c) Compute $\text{Cov}(u, s^*)$ and say whether the LM error is classical
- ▶ (d) Compare λ with the FinBERT value of A1: which classifier attenuates more, and by what ratio?

□ What to report

- ▶ The four rates, λ , $1/\lambda$ and the ratio of the two factors
- ▶ Interpretation: is the weaker same-day LM slope of Lecture 15 (15.5 bp against 39.2 bp for FinBERT) evidence of a weaker market reaction to LM-negative news? Model: A1

A3: Paired Accuracy Difference: Variance, Exact Test, Power (task 1/2) [Solved]

Task

▣ Context and assumptions

- ▶ Data: Twitter Financial News, validation set of $N = 2\,388$ headlines published in 2022, each labelled negative, neutral or positive by annotators; the human label is treated as the truth
- ▶ Classifier A = LM dictionary, classifier B = FinBERT; discordant counts $n_{01} = 539$ (A wrong, B right) and $n_{10} = 252$ (A right, B wrong); headlines are independent
- ▶ For headline i , $D_i = 1$ if A is wrong and B right, $D_i = -1$ if A is right and B wrong, $D_i = 0$ otherwise; the accuracy difference is $d = ED_i = \text{acc}_B - \text{acc}_A$

A3: Paired Accuracy Difference: Variance, Exact Test, Power (task 2/2) [Solved]

Task

What to do

- ▶ (a) Show that $\hat{d} = (n_{01} - n_{10})/n$ and derive $\text{Var}(\hat{d}) = (q - d^2)/n$, where $q = P(D_i \neq 0)$ is estimated by $(n_{01} + n_{10})/n$
- ▶ (b) Compute $\hat{d}$, its standard error, the 95% CI $\hat{d} \pm 1.96 \text{ SE}$ and $z = \hat{d}/\text{SE}$
- ▶ (c) Run the exact McNemar test [McNemar \(1947\)](#): under $H_0 : d = 0$, $n_{01} \sim \text{Bin}(n_{01} + n_{10}, 0.5)$; give the two-sided p -value and the decision at $\alpha = 5\%$
- ▶ (d) Approximate the power at $n = 2388$ if the true d equals $\hat{d}$, $\text{power} \approx \Phi(|d|\sqrt{n}/\sqrt{q - d^2} - 1.96)$, and the n needed for 80% power, $n_{80} = (1.96 + 0.84)^2(q - d^2)/d^2$

What to report

- ▶ $\hat{d}$, SE, CI, z , exact p , the decision, the power and n_{80} ; compare the CI with the paired bootstrap CI of B1
- ▶ Interpretation: which headlines determine the variance of $\hat{d}$?

A3: Solution (1/2) [Solved]

Solution

- (a) $\hat{d} = \frac{1}{n} \sum_i D_i = (n_{01} - n_{10})/n$, because 539 terms equal +1, 252 equal -1 and the rest are 0
- $D_i^2 = 1$ exactly when one classifier is right and the other wrong (two wrong but different labels give $D_i = 0$), so $ED_i^2 = q$ and $\text{Var}(D_i) = q - d^2$; with independent headlines, $\text{Var}(\hat{d}) = (q - d^2)/n$
- (b) $\hat{d} = (539 - 252)/2\,388 = 0.1202$, $\hat{q} = 791/2\,388 = 0.3312$
- $SE = \sqrt{(0.3312 - 0.01444)/2\,388} = 1.15 \text{ pp}$; CI 95% [9.76; 14.28] pp, close to the bootstrap CI [+9.8; +14.2] of B1; $z = 10.43$

A3: Solution (2/2) [Solved]

Solution

- ▣ (c) Under H_0 we expect 395.5 of the 791 discordant headlines on each side; observing 539 gives an exact two-sided $p < 0.001$
- ▣ Decision: reject H_0 at 5%; FinBERT is more accurate than LM on these headlines
- ▣ (d) $|d|\sqrt{n}/\sqrt{q-d^2} - 1.96 = +8.47$, so the power is $\Phi(+8.47) \approx 100\%$; $n_{80} = 7.84 \times (0.3312 - 0.01444)/0.01444 \approx 170$ headlines
- ▣ Interpretation: only the discordant headlines (q) enter the variance; a 12-point gap is detectable with a few hundred headlines

A4: Is “No Difference” a Power Problem? [Proposed]

Solution discussed in class

Task

□ Context and assumptions

- ▶ Data: Twitter Financial News, validation set of $N = 2\,388$ headlines published in 2022, each labelled negative, neutral or positive by annotators; the human label is treated as the truth
- ▶ Classifier A = FinBERT, classifier B = Qwen2.5-7B zero-shot (prompt P1); $n_{01} = 395$ (A wrong, B right), $n_{10} = 355$ (A right, B wrong)
- ▶ For headline i , $D_i = 1$ if A is wrong and B right, $D_i = -1$ if A is right and B wrong, $D_i = 0$ otherwise; the accuracy difference is $d = ED_i = \text{acc}_B - \text{acc}_A$

□ What to do

- ▶ (a) Compute $\hat{d}$, $\hat{q}$, SE, the 95% CI and z with the formulas of A3
- ▶ (b) Run the exact McNemar test at $\alpha = 5\%$ and state the decision
- ▶ (c) Compute the power at $n = 2\,388$ if the true d equals $\hat{d}$, and n_{80}
- ▶ (d) Use the CI to state which differences in accuracy the data rule out

□ What to report

- ▶ All the numbers of (a)–(c); interpretation: may you conclude that the two models are equally accurate? Model: A3

A5 (1/2): From Label Probabilities to a Score [Solved]

Task

□ Context and assumptions

- ▶ Qwen2.5-7B, zero-shot with prompt P1, one Twitter validation headline that the annotators labelled **neutral**
- ▶ Answer-word probabilities (rounded): $p_{\text{neg}} = 0.608$, $p_{\text{neu}} = 0.392$, $p_{\text{pos}} = 0.000$ (exactly 1.0×10^{-12}); they come from logits by $p_k = e^{z_k/T} / \sum_j e^{z_j/T}$ with $T = 1$

□ What to do

- ▶ (a) Give the predicted label $\arg \max_k p_k$ and the net score $s = p_{\text{pos}} - p_{\text{neg}}$, and say whether the label is right
- ▶ (b) Compute the logit gap $z_{\text{neu}} - z_{\text{neg}} = \ln(p_{\text{neu}}/p_{\text{neg}})$
- ▶ (c) Compute the probabilities, the label and the score at $T = 2$, using $p_k(T) \propto p_k^{1/T}$

□ What to report

- ▶ The label, s , the logit gap and $p(T)$, $s(T)$; interpretation: what happens to a trading rule that uses a threshold on s when T changes?

A5 (1/2): Solution [Solved]

Solution

- ▣ (a) The largest probability is $p_{\text{neg}} = 0.608$: the label is negative, which is wrong (the truth is neutral)
- ▣ $s = p_{\text{pos}} - p_{\text{neg}} = 0.000 - 0.608 = -0.608$: a strongly negative score for a neutral headline
- ▣ (b) $z_{\text{neu}} - z_{\text{neg}} = \ln(0.392/0.608) = -0.437$: neutral is behind by less than half a logit
- ▣ (c) With $T = 2$, $p_k(T) \propto \sqrt{p_k}$: $\sqrt{p} = (0.780, 0.626, 1.0 \times 10^{-6})$ with sum 1.406, so $p(T) = (0.554, 0.446, 0.000)$
- ▣ The label stays negative and $s(T) = -0.554$; interpretation: temperature keeps the ranking but pulls the score towards 0, so a rule such as $|s| > 0.5$ selects different headlines

A5 (2/2): Fitting the Temperature [Solved]

Task

▣ Context and assumptions

- ▶ Qwen2.5-7B probabilities on a random half of the Twitter validation set ($n = 1\,194$ headlines, seed 15); logits $z_{ik} = \ln p_{ik}$, true label y_i
- ▶ Rescaled probabilities $p_{ik}(T) = e^{z_{ik}/T} / \sum_l e^{z_{il}/T}$, $T > 0$; loss $\text{NLL}(T) = -\sum_i \ln p_{iy_i}(T)$ Guo et al. (2017)

▣ What to do

- ▶ (a) Explain why the predicted labels and the accuracy do not depend on T , while the Brier score does
- ▶ (b) Derive $\text{NLL}'(T)$ and write the first-order condition for the optimal T
- ▶ (c) Evaluate the average bracket of the condition at $T = 1$, then solve for T^* with a root finder (Brent's method on $[0.5, 50]$)
- ▶ (d) On the other half, compute the ECE (10 equal-width bins of the top probability on $[1/3, 1]$) and the Brier score $\frac{1}{n} \sum_i \sum_k (p_{ik}(T) - 1\{y_i = k\})^2$ (summed over the 3 classes) before and after scaling

▣ What to report

- ▶ T^* , the sign at $T = 1$, ECE and Brier before and after; interpretation: is the LLM over- or under-confident?

A5 (2/2): Solution [Solved]

Solution

- (a) $z \mapsto z/T$ is increasing for $T > 0$, so the order of z_{i1}, z_{i2}, z_{i3} and hence $\arg \max$ and accuracy do not change; the probabilities change, and with them Brier and log loss
- (b) $\ln p_{iy}(T) = z_{iy}/T - \ln \sum_k e^{z_{ik}/T}$, so $\frac{d}{dT} \ln p_{iy}(T) = -\frac{z_{iy}}{T^2} + \frac{1}{T^2} \sum_k p_{ik}(T) z_{ik}$
- $\text{NLL}'(T) = \frac{1}{T^2} \sum_i (z_{iy_i} - \sum_k p_{ik}(T) z_{ik}) = 0$: on average the true-class logit must equal its expected value under the model
- (c) At $T = 1$ the average bracket is $-2.47 < 0$, so NLL decreases as T grows; the root is $T^* = 8.43$
- (d) ECE falls from 22.6% to 4.3% and the Brier score from 0.468 to 0.366; interpretation: the raw LLM probabilities are heavily overconfident

A6: Fleiss' κ and One Dawid–Skene Step (task 1/2) [Proposed]

Solution discussed in class

Task

□ Context and assumptions

- ▶ Illustrative table (numbers made up for the exercise): 6 sentences, $m = 5$ raters each, votes for negative, neutral, positive
- ▶ Fleiss' κ [Fleiss \(1971\)](#): $P_i = \sum_j n_{ij}(n_{ij} - 1)/[m(m - 1)]$, $\bar{P} = \frac{1}{N} \sum_i P_i$, $p_j =$ share of all votes for j , $\bar{P}_e = \sum_j p_j^2$,
 $\kappa = (\bar{P} - \bar{P}_e)/(1 - \bar{P}_e)$

Sentence	neg	neu	pos
1	5	0	0
2	0	5	0
3	0	1	4
4	1	3	1
5	2	3	0
6	0	2	3

A6: Fleiss' κ and One Dawid–Skene Step (task 2/2) [Proposed]

Solution discussed in class

Task

□ What to do

- ▶ (a) Compute P_i for every sentence and $\bar{P}$
- ▶ (b) Compute p_j , $\bar{P}_e$ and κ
- ▶ (c) E-step of [Dawid & Skene \(1979\)](#): assume every rater gives the true class with probability 0.8 and each other class with probability 0.1, and a uniform prior; compute $P(\text{true} = j \mid \text{votes}) \propto \prod_l \pi_{jl}^{n_{jl}}$ for sentences 4 and 5
- ▶ (d) Compare the posterior with the vote shares and describe the M-step

□ What to report

- ▶ κ and the two posteriors; interpretation: which sentences would you drop from a test set, and why? Model: A1 (misclassification probabilities)

A7: Break-Even Cost with a Standard Error [Solved]

Task

▣ Context and assumptions

- ▶ Strategy of Lecture 15 on the 16 FNSPID stocks, 3568 trading days: at the close of $d + 1$ take $\text{sign}(s_{i,d})$ in every stock with news on day d , hedged with SPY, close one day later; four trades on each day with positions
- ▶ FinBERT signal: mean gross return over all days $\hat{\mu} = 3.27$ bp, Newey–West $t = 1.27$ Newey & West (1987) with $L = 8$ lags ($L = \lfloor 4(n/100)^{2/9} \rfloor$), share of days with positions $a = 0.962$, gross Sharpe $\sqrt{252} \hat{\mu} / \sigma = 0.36$

▣ What to do

- ▶ (a) Write the mean net daily return and the annualised net Sharpe ratio as functions of the cost c per trade, assuming the costs barely change the volatility σ
- ▶ (b) Derive the break-even cost c^* at which the mean net return is zero
- ▶ (c) Compute $\text{SE}(\hat{\mu}) = \hat{\mu} / t$, then $\text{SE}(c^*)$ by the delta method and the 95% CI $c^* \pm 1.96 \text{SE}$
- ▶ (d) Recover σ from the gross Sharpe ratio and compute the net Sharpe ratio at $c = 5$ bp

▣ What to report

- ▶ c^* , SE, CI and the net Sharpe ratio; interpretation: is a realistic cost of 5 bp per trade inside the CI of c^* ?

A7: Solution [Solved]

Solution

- (a) On a fraction a of the days we pay $4c$, so $\mu_{\text{net}} = \mu - 4ac$ and $SR_{\text{net}} = \sqrt{252}(\mu - 4ac)/\sigma$
- (b) $\mu - 4ac^* = 0$ gives $c^* = \mu/(4a) = 3.27/3.85 = 0.85$ bp per trade
- (c) $SE(\hat{\mu}) = 3.27/1.27 = 2.57$ bp; treating a as fixed (the activity schedule is set by the news calendar), c^* is linear in $\hat{\mu}$, so $SE(c^*) = 2.57/3.85 = 0.67$ bp and the CI is $[-0.46; 2.16]$ bp
- (d) $\sigma = \sqrt{252} \hat{\mu} / SR = \sqrt{252} \times 3.27 / 0.36 = 143$ bp; $SR_{\text{net}} = \sqrt{252}(3.27 - 3.85 \times 5) / 143 = -1.77$
- Interpretation: 5 bp lies far above the upper bound of the CI; with daily turnover the strategy cannot survive realistic costs

A8: The Qwen Strategy after Costs [Proposed]

Solution discussed in class

Task

□ Context and assumptions

- ▶ Strategy of Lecture 15 on the 16 FNSPID stocks, 3568 trading days: at the close of $d + 1$ take $\text{sign}(s_{i,d})$ in every stock with news on day d , hedged with SPY, close one day later; four trades on each day with positions
- ▶ Qwen2.5-7B signal: $\hat{\mu} = 5.64$ bp, Newey–West $t = 2.26$ ($L = 8$), $a = 0.950$, gross Sharpe 0.63

□ What to do

- ▶ (b) Derive the break-even cost c^* at which the mean net return is zero
- ▶ (c) Compute $\text{SE}(\hat{\mu}) = \hat{\mu}/t$, then $\text{SE}(c^*)$ by the delta method and the 95% CI $c^* \pm 1.96 \text{ SE}$
- ▶ (d) Recover σ from the gross Sharpe ratio and compute the net Sharpe ratio at $c = 5$ bp
- ▶ (e) Lecture 15 reports a break-even cost of 1.5 bp and a net Sharpe ratio of -1.48 : decide whether this is a copy error

□ What to report

- ▶ c^* , SE, CI, net Sharpe; interpretation: does the significant gross mean make the strategy tradable? Model: A7

B1: Is FinBERT Better than LM? [Solved]

Task

▣ Context and assumptions

- ▶ Data: Twitter Financial News, validation set of $N = 2\,388$ headlines published in 2022, each labelled negative, neutral or positive by annotators; the human label is treated as the truth
- ▶ Classifier A = LM dictionary (label = sign of the tone), classifier B = FinBERT (label = largest probability); code: `b_pairs` in the notebook

▣ What to do

- ▶ (a) Compute the accuracy of each classifier with a percentile bootstrap 95% CI: resample the headlines with replacement 2 000 times (seed 0) and take the 2.5% and 97.5% quantiles
- ▶ (b) Compute the difference $B - A$ with a paired bootstrap CI: the same resampled headlines for both classifiers in each of the 2 000 draws
- ▶ (c) Count n_{01} and n_{10} , run the exact McNemar test and the asymptotic version $(n_{01} - n_{10})^2 / (n_{01} + n_{10})$ against χ_1^2 (critical value 3.84) at $\alpha = 5\%$

▣ What to report

- ▶ A table with the accuracies, CIs and the difference; n_{01} , n_{10} , p -values and the decision
- ▶ Interpretation: why is the paired test more powerful than comparing the two intervals, which headlines carry no information, and is the gap useful for trading?

B1: Solution (1/2) [Solved]

	Accuracy (%)	CI 95%
LM	60.5	[58.5; 62.4]
FinBERT	72.5	[70.7; 74.4]
Difference (pp)	+12.0	[+9.8; +14.2]

Solution

- Step 1: labels of both classifiers on the 2388 headlines; step 2: 2000 bootstrap resamples of the headline indices; step 3: accuracies and their difference in each resample
- (a)–(b) The intervals of the two classifiers do not overlap and the paired CI of the difference excludes 0

B1: Solution (2/2) [Solved]

Solution

- (c) $n_{01} = 539$ (LM wrong, FinBERT right), $n_{10} = 252$; under H_0 we expect 395.5 each; exact binomial $p < 0.001$
- Asymptotic: $(539 - 252)^2 / 791 = 104.1 > 3.84$; decision: reject equal accuracy at 5%
- Pairing removes the variation of headline difficulty: headlines that both get right or both get wrong cancel out and carry no information
- Interpretation: FinBERT is clearly better than the dictionary outside its own training data; a 12-point gain in labelling accuracy does not yet say anything about returns (B5)

B2: Is a 7-Billion-Parameter LLM Better than FinBERT? [Proposed]

Solution discussed in class

Task

▣ Context and assumptions

- ▶ Data: Twitter Financial News, validation set of $N = 2\,388$ headlines published in 2022, each labelled negative, neutral or positive by annotators; the human label is treated as the truth
- ▶ Classifier A = FinBERT, classifier B = Qwen2.5-7B zero-shot with prompt P1; neither used any Twitter label

▣ What to do

- ▶ (a) Compute the accuracy of each classifier with a percentile bootstrap 95% CI: resample the headlines with replacement 2 000 times (seed 0) and take the 2.5% and 97.5% quantiles
- ▶ (b) Compute the difference $B - A$ with a paired bootstrap CI: the same resampled headlines for both classifiers in each of the 2 000 draws
- ▶ (c) Count n_{01} and n_{10} , run the exact McNemar test and the asymptotic version $(n_{01} - n_{10})^2 / (n_{01} + n_{10})$ against χ^2_1 (critical value 3.84) at $\alpha = 5\%$

▣ What to report

- ▶ The same table and tests as in B1; interpretation: do the two intervals overlap, does that settle the question, and what would you still gain from a much larger and slower model? Model: B1

B3: Training-Data Leakage [Solved]

Task

▣ Context and assumptions

- ▶ Financial PhraseBank [Malo et al. \(2014\)](#): 4 846 news sentences (2 264 with full annotator agreement); FinBERT was fine-tuned on a training split of these sentences (3 101 of 4 845 in [Araci \(2019\)](#))
- ▶ Twitter validation set, 2 388 headlines, unseen by FinBERT; four classifiers: FinBERT, Qwen2.5-7B zero-shot, LM, MiniLM embeddings + logistic regression trained on the Twitter training set [Wang et al. \(2020\)](#)

▣ What to do

- ▶ (a) Compute accuracy (with a 2 000-draw bootstrap CI) and macro-F1 of each classifier on PhraseBank with full agreement, on all of PhraseBank and on Twitter
- ▶ (b) Compute the drop from all of PhraseBank to Twitter for each classifier
- ▶ (c) Compute the difference in differences FinBERT minus Qwen2.5-7B and its bootstrap CI, resampling each data set independently 2 000 times

▣ What to report

- ▶ A table of accuracies, drops and macro-F1; the difference in differences with its CI
- ▶ Interpretation: how do you separate “harder data” from “memory of the training set”, and why does MiniLM move the other way?

B3: Solution [Solved]

Accuracy (%)	PhraseBank, 100% agr.	PhraseBank, all	Twitter	Drop (pp)	Macro-F1, Twitter
FinBERT	97.2	89.0	72.5	16.4	66.8
Qwen2.5-7B	93.6	81.4	74.2	7.2	72.6
LM	65.0	62.5	60.5	2.0	46.2
MiniLM + log. reg.	81.4	70.6	78.6	−8.0	69.9

Solution

- (b) Drop = all of PhraseBank minus Twitter; FinBERT CI on PhraseBank [88.1; 89.8], on Twitter [70.7; 74.4]
- (c) $(89.0 - 72.5) - (81.4 - 74.2) = 16.4 - 7.2 = 9.2$ pp, bootstrap CI [6.7; 11.9]: it excludes 0
- Interpretation: Qwen2.5-7B was not fine-tuned on PhraseBank labels, so its drop is the benchmark for the harder data; if both would lose the same without training overlap (an untested assumption), FinBERT's extra 9.2 points come from overlap; the CI covers sampling noise only; MiniLM is trained on Twitter, so it is better at home

B4: How Many Labels Do You Need? [Proposed]

Solution discussed in class

Task

▣ Context and assumptions

- ▶ Twitter Financial News: training set of 9 543 labelled headlines, validation set of 2 388; benchmark FinBERT at 72.5% accuracy, which uses no Twitter label
- ▶ Two supervised models: logistic regression ($C = 4$) on 384-dimensional MiniLM embeddings, and logistic regression ($C = 8$) on TF-IDF features of words and word pairs

▣ What to do

- ▶ (a) For $n \in \{100, 200, 400, \dots, 6\,400, 9\,543\}$ draw 10 random training subsets of size n (one for the full set), fit both models and compute the mean validation accuracy
- ▶ (b) Find the smallest n at which each model beats FinBERT on average
- ▶ (c) Report the standard deviation of the embedding accuracy across the 10 repetitions at $n = 100$

▣ What to report

- ▶ The learning-curve table; interpretation: why do embeddings start higher but grow more slowly, and how many texts would you label for Romanian stock news? Model: B3

B5: Reaction or Prediction, and Which Standard Errors? [Solved]

Task

▣ Context and assumptions

- ▶ Panel of 18 282 stock-days with news on 16 FNSPID stocks, 3 434 trading days, 27.10.2009–18.12.2023; daily score = mean of the headline scores, standardised over the panel (z)
- ▶ Returns: daily adjusted closes of the stocks and of SPY from data/market; excess return r^e = stock minus SPY, in %; day d = first trading day on or after the headline date, so $d + 2$ is the first return surely after publication

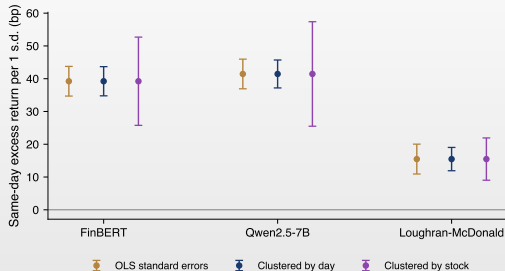
▣ What to do

- ▶ (a) Estimate by OLS $r_{i,d+h}^e = a + b z_{i,d} + \varepsilon$ for FinBERT, Qwen2.5-7B and LM and $h = 0, 1, 2$ (nine regressions); report b in bp per one standard deviation of the score
- ▶ (b) Compute three standard errors: OLS, clustered by day ($G = 3\,434$) and clustered by stock ($G = 16$), with the sandwich estimator and the small-sample factor $\frac{G}{G-1} \frac{n-1}{n-2}$ [Petersen \(2009\)](#)
- ▶ (c) Compute the t -statistics and compare them with 1.96 and with the t_{15} critical value 2.13 for the stock clustering

▣ What to report

- ▶ A chart of the $h = 0$ slopes with the three intervals and a table of all nine regressions
- ▶ Interpretation: why do $h = 0$ and $h = 2$ differ, which clustering matters, and can you trust a clustered SE with 16 clusters?

B5: Solution (1/3) [Solved]



Solution

- Same-day slope b (bp per one standard deviation of score) with 95% intervals from the three standard errors: the stock-clustered interval is much wider

B5: Solution (2/3) [Solved]

Score	h	b (bp)	SE OLS	SE day	SE stock	t day	t stock
FinBERT	0	+39.2	2.3	2.3	6.9	+17.27	+5.71
FinBERT	2	-0.5	2.2	2.2	1.5	-0.25	-0.35
Qwen2.5-7B	0	+41.4	2.3	2.2	8.1	+19.07	+5.10
Qwen2.5-7B	2	+2.3	2.2	1.9	1.0	+1.19	+2.41
LM	0	+15.5	2.3	1.8	3.3	+8.53	+4.70
LM	2	-0.0	2.2	1.7	0.9	-0.01	-0.03

Solution

- $h = 0$: large under every standard error (t by stock above 4.7): headlines describe the move of the day, often the move itself ("shares jump")
- $h = 2$: FinBERT -0.5 bp ($t = -0.25$ by day): the tradable forecast is not there

B5: Solution (3/3) [Solved]

Solution

- Clustering by day is close to OLS: with only 5.3 stocks per day there is little common noise to absorb
- Clustering by stock multiplies the FinBERT SE at $h = 0$ by 3.0: the slope differs across stocks, and each stock's errors are correlated over time
- With 16 clusters the clustered SE is itself noisy: Qwen at $h = 2$ has $t = +2.41$ by stock, $p = 0.029$ with t_{15} , but 0.061 by wild cluster bootstrap [Cameron, Gelbach & Miller \(2008\)](#) and 0.37 after Holm over the nine regressions (B10)
- Interpretation: sentiment explains the news day, not the tradable days; with 16 stocks, inference must use few-cluster methods

B6: Event Study with the Loughran–McDonald Tone (task 1/2) [Proposed]

Solution discussed in class

Task

□ Context and assumptions

- ▶ Panel of 18 282 stock-days with news on 16 FNSPID stocks, 3 434 trading days, 27.10.2009–18.12.2023; daily score = mean of the headline scores, standardised over the panel (z)
- ▶ Returns: daily adjusted closes of the stocks and of SPY from data/market; excess return r^e = stock minus SPY, in %; day d = first trading day on or after the headline date, so $d + 2$ is the first return surely after publication

B6: Event Study with the Loughran–McDonald Tone (task 2/2) [Proposed]

Solution discussed in class

Task

What to do

- ▶ (a) The daily LM tone is exactly 0 on 48% of stock-days, so both tercile cut-offs are 0: form the disjoint groups $\text{tone} < 0$ (5 196 stock-days) and $\text{tone} > 0$ (4 288), dropping the zeros
- ▶ (b) For each date, average across that date's stock-days the excess returns on days $-5, \dots, +5$ around the news day
- ▶ (c) Compute the mean over dates of the CAR (cumulative abnormal return) for $[-5, -1]$, day 0, day 1 and $[+2, +5]$, with Newey–West t (lags $\lfloor 4(n/100)^{2/9} \rfloor$)
- ▶ (d) Repeat (b)–(c) for the bottom and top terciles of the FinBERT score (no ties there)

What to report

- ▶ A table of the four windows for both scores and both groups; interpretation: does LM separate winners from losers on day 0 as well as FinBERT, and why is the post-event return positive for both groups [Tetlock \(2007\)](#)? Model: B5

B7: Does the Model Remember the Market? [Solved]

Task

□ Context and assumptions

- ▶ S&P 500 monthly closes from data/market (GSPC.INDX), January 2000 to August 2026; a month is “up” if its return is positive
- ▶ Qwen2.5 1.5B, 7B and 14B, weights published on 19 September 2024 [Qwen Team \(2024\)](#), asked without any context “Did the S&P 500 index rise or fall during ⟨month year⟩?”; score $P(\text{rise})$
- ▶ Before: 296 months to August 2024 (185 up); after: 23 months from October 2024 (13 up)

□ What to do

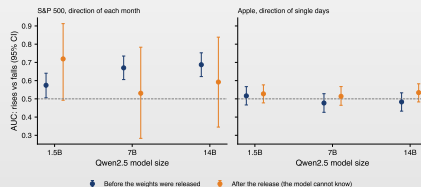
- ▶ (a) Compute the $\text{AUC} = U/(n_{\text{up}}n_{\text{down}})$ (Mann–Whitney U) for each model and period, with a 2000-draw bootstrap CI over months
- ▶ (b) Run a permutation test for 14B before the release: shuffle the up/down labels 2000 times; p = share of shuffled AUCs at least as large as the observed one
- ▶ (c) Test $\text{AUC}_{\text{before}} = \text{AUC}_{\text{after}}$ with DeLong variances [DeLong et al. \(1988\)](#) (independent samples) at 5%, and compute the smallest drop detectable with 80% power

□ What to report

- ▶ A table of AUCs with CIs and a chart; the permutation p , the DeLong z and p , the detectable drop
- ▶ Interpretation: what does an AUC above 0.5 mean when the model sees no data, and what does it imply for backtests on old news [Glasserman & Lin \(2023\)](#)?

B7: Solution (1/2) [Solved]

Qwen2.5	AUC before	CI 95%	AUC after	CI 95%	$\bar{P}(\text{rise})$ up / down
1.5B	0.57	[0.51; 0.64]	0.72	[0.49; 0.91]	0.36 / 0.30
7B	0.67	[0.61; 0.73]	0.53	[0.28; 0.78]	0.73 / 0.43
14B	0.69	[0.62; 0.75]	0.59	[0.35; 0.84]	0.81 / 0.54



Solution

- (a) 7B and 14B rank up months above down months well above chance before the release; the 1.5B model barely

B7: Solution (2/2) [Solved]

Solution

- (b) Permutation p for 14B before the release < 0.001 (2000 shuffles): reject “no knowledge” at 5%
- (c) DeLong: SE before 0.032, after 0.126; $z = 0.73$, $p = 0.47$: do not reject equality
- Under H_0 the post-release SE is 0.124, so the smallest drop detectable with 80% power is $2.80 \times \sqrt{SE_{\text{before}}^2 + SE_0^2} = 0.36$; about 101 post-release months would be needed
- Interpretation: no data were given, so the discrimination most plausibly comes from the training text (outcomes or the regimes around them; months treated as exchangeable in the bootstrap and permutation); the post-release sample is far too short to show that the memory is gone, and a backtest on pre-release news mixes forecasting with remembering

B8: Prompt Sensitivity [Proposed]

Solution discussed in class

Task

▣ Context and assumptions

- ▶ Data: Twitter Financial News, validation set of $N = 2\,388$ headlines published in 2022, each labelled negative, neutral or positive by annotators; the human label is treated as the truth
- ▶ Qwen2.5-7B and 14B with three prompts that ask the same question: P1 “classify the sentiment . . . for investors”, P2 “good, bad or neutral for the stock price”, P3 “what is the sentiment of this text”

▣ What to do

- ▶ (a) Compute the accuracy with a 2 000-draw bootstrap CI and the share of “neutral” answers for each prompt and model
- ▶ (b) Run the exact McNemar test of P1 against P3 for each model at 5%
- ▶ (c) Compute the share of headlines that get the same label under all three prompts

▣ What to report

- ▶ A table with the three prompts and the test; interpretation: if you picked the best prompt on this set, what would you be doing, and why does the best prompt also answer “neutral” most often? Model: B1

B9: Prediction-Powered Inference with a Few Human Labels [Solved]

Task

Context and assumptions

- ▶ Data: Twitter Financial News, validation set of $N = 2\,388$ headlines published in 2022, each labelled negative, neutral or positive by annotators; the human label is treated as the truth
- ▶ Target $\theta =$ mean human net tone $(-1, 0, 1)$ of the headline population, represented by the 2 388 headlines; model score $f =$ net score $P(\text{pos}) - P(\text{neg})$ of FinBERT or of Qwen2.5-7B
- ▶ Design: n headlines drawn at random reveal their human label Y ; the other $N - n$ keep only f ; f is fixed (not trained on these headlines), so the two sets are independent samples

What to do

- ▶ (a) For one draw with $n = 200$, compute the labels-only estimate $\bar{Y}_n \pm 1.96 s_Y / \sqrt{n}$
- ▶ (b) Compute PPI [Angelopoulos et al. \(2023\)](#): $\hat{\theta} = \frac{1}{N-n} \sum_{\text{unlab.}} f - \frac{1}{n} \sum_{\text{lab.}} (f - Y)$ with SE $= \sqrt{\hat{V}(f)/(N-n) + \hat{V}(f - Y)/n}$, for both models
- ▶ (c) For $n = 100, 200, 400, 800$, draw 2 000 times n labelled and $N - n$ unlabelled headlines independently, with replacement (the 2 388 act as the population), and record the coverage of θ and the mean width of each 95% CI

What to report

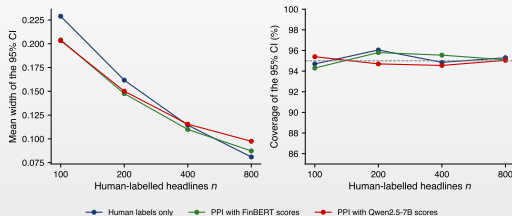
- ▶ The one-draw estimates, a coverage and width table or chart; interpretation: why is the plain model mean $\frac{1}{N} \sum f$ not an estimate of θ , when is PPI narrower, and how would you use it for the FNSPID regressions?

B9: Solution (1/2) [Solved]

Solution

- ▣ True $\theta = 0.054$; plain model means: FinBERT -0.014 (wrong sign), Qwen $+0.101$: the model bias does not shrink as N grows
- ▣ (a) One draw, $n = 200$: labels only $0.010 \pm 1.96 \times 0.042$
- ▣ (b) PPI-FinBERT: rectifier $\frac{1}{n} \sum (f - Y) = -0.064$, so $\hat{\theta} = 0.054 \pm 1.96 \times 0.036$; PPI-Qwen: rectifier -0.012 , $\hat{\theta} = 0.123 \pm 1.96 \times 0.039$
- ▣ The rectifier removes the bias in expectation, but in one draw it can move an estimate away from θ (Qwen here); all three intervals contain the true value in this draw

B9: Solution (2/2) [Solved]



Solution

- (c) $n = 100$: coverage 94.7%, 94.3%, 95.4%, width 0.229, 0.204, 0.203 (labels, PPI-FinBERT, PPI-Qwen)
- PPI is narrower when $\text{Var}(f)/(N - n) + \text{Var}(f - Y)/n < \text{Var}(Y)/n$; for large N the gain is $1 - \text{Var}(f - Y)/\text{Var}(Y)$: 24% (FinBERT), 27% (Qwen); at $n = 800$ only $N - n = 1\,588$ texts are unlabelled and PPI is wider (0.087 vs 0.081); coverage stays near 95%
- Interpretation: for FNSPID, label all headlines of a random sample of stock-days and rectify every OLS moment condition that contains the score (C1, part (e))

B10: Wild Cluster Bootstrap with 16 Stocks [Proposed]

Solution discussed in class

Task

Context and assumptions

- ▶ Panel of 18 282 stock-days with news on 16 FNSPID stocks, 3 434 trading days, 27.10.2009–18.12.2023; daily score = mean of the headline scores, standardised over the panel (z)
- ▶ The nine regressions of B5 (three scores $\times h = 0, 1, 2$), with standard errors clustered by stock ($G = 16$); one-way clustering assumes independent stocks

What to do

- ▶ (a) Compute the p -value of each slope with t_{15} critical values
- ▶ (b) Run the wild cluster bootstrap with the null imposed [Cameron, Gelbach & Miller \(2008\)](#): $y_i^* = \bar{y} + w_{g(i)}\tilde{\varepsilon}_i$ with restricted residuals $\tilde{\varepsilon}_i = y_i - \bar{y}$, Rademacher weights $w_g = \pm 1$, 9 999 draws; $p = P^*(|t^*| \geq |t|)$
- ▶ (c) Compute t with two-way clustering by day and stock, $\hat{V} = \hat{V}_{\text{day}} + \hat{V}_{\text{stock}} - \hat{V}_{\text{day} \cap \text{stock}}$
- ▶ (d) Adjust the nine bootstrap p -values with Holm's step-down procedure at 5%

What to report

- ▶ A table with t , the three p -values and the two-way t ; interpretation: why is $N(0, 1)$ a poor reference for a t built from 16 clusters, and does Qwen at $h = 2$ survive? Model: B5

B11: Specification Curve for the Sentiment Strategy [Proposed]

Solution discussed in class

Task

□ Context and assumptions

- ▶ Strategy of Lecture 15 on the 16 FNSPID stocks, 3568 trading days: at the close of $d + 1$ take $\text{sign}(s_{i,d})$ in every stock with news on day d , hedged with SPY, close one day later; four trades on each day with positions
- ▶ Grid of Lecture 15: three scores (LM, FinBERT, Qwen) $\times$ thresholds $|s| > b$, $b \in \{0, 0.25, 0.5\}$, $\times$ return day $d + 2, \dots, d + 5$ (each a one-day position opened at the previous close): 36 strategies

□ What to do

- ▶ (a) Compute the mean daily return and its Newey–West t for each of the 36 strategies, and plot the t -statistics sorted [Simonsohn, Simmons & Nelson \(2020\)](#)
- ▶ (b) Simulate the joint null of no predictability: flip the signs of the positions at random in blocks of 10 consecutive signal days (dependent wild bootstrap [Shao \(2010\)](#)), the same flips for all 36 strategies, 2000 times
- ▶ (c) Compute the joint p -values of the number of $t > 1.96$ and of the median t

□ What to report

- ▶ The sorted curve and the two joint p -values; interpretation: why must the flip be shared by all specifications, and what would a “best specification” report hide? Model: B7 (a randomisation test)

C1: Does Headline Sentiment Pay after Costs? (task 1/2) [Proposed]

Solution discussed in class

Task

▣ Context and assumptions

- ▶ Strategy of Lecture 15 on the 16 FNSPID stocks, 3568 trading days: at the close of $d + 1$ take $\text{sign}(s_{i,d})$ in every stock with news on day d , hedged with SPY, close one day later; four trades on each day with positions
- ▶ Positions are hedged with SPY and equally weighted; days without news: no position and return 0; costs of 5 bp per trade, four trades on each day with positions
- ▶ Scores: FinBERT, Qwen2.5-7B and LM, from the saved tables of Lecture 15; returns from data/market

C1: Does Headline Sentiment Pay after Costs? (task 2/2) [Proposed]

Solution discussed in class

Task

What to do

- ▶ (a) Compute the mean daily gross return, its Newey–West t and a moving-block bootstrap 95% CI (blocks of 10 days, 2 000 draws) for the FinBERT signal
- ▶ (b) Compute the annualised Sharpe ratio before and after costs, the break-even cost and the Sharpe ratio for costs of 0, 1, 2, 5 and 10 bp
- ▶ (c) Restrict to strong news, $|s_{i,d}| > 0.5$ (exploratory), and (d) repeat everything for Qwen2.5-7B and LM; the primary family, fixed in advance, is the three unfiltered strategies (Holm over them); if you select among all six variants, adjust over all six
- ▶ (e) Correct the same-day slope: label all headlines of a random sample of stock-days (true daily tone s^*); either divide b by $\rho(s^*, \hat{s})$ estimated there (the lecture correction for a standardised score; A1 applies only to a dummy), or rectify both OLS moments $E[(1, s^*)^\top (r - a - \beta s^*)] = 0$ with the score (B9, lecture); bootstrap over days
- ▶ (f) Report headlines after the Qwen2.5 release separately once available

What to report

- ▶ A two-page note with a results table and charts, and a Quantlet folder (code, Metainfo, charts); discuss timing, look-ahead of the LLM (B7), survivorship of the 16 stocks and crowding [Lopez-Lira & Tang \(2026\)](#)

C2: Critique an AI Answer (1/3) [Proposed]

Solution discussed in class

- **Student prompt:** “I have FNSPID headlines for 16 US stocks, 2009–2023, and daily excess returns vs SPY. Check that Qwen2.5-7B reads sentiment, then test whether its headline sentiment forecasts returns out of sample.”
- **Answer of the AI assistant** (qwen_label, qwen_score: the zero-shot scorer of Lecture 15)

```
import pandas as pd, statsmodels.formula.api as smf
from sklearn.metrics import accuracy_score
tw['pred'] = [qwen_label(t) for t in tw['text']] # Twitter validation set
acc = accuracy_score(tw['label'], tw['pred'])
news['day'] = first_trading_day_on_or_after(news['date'])
news['s'] = [qwen_score(t) for t in news['title']] # P(pos) - P(neg)
d = news.groupby(['ticker', 'day'])['s'].mean().reset_index()
d['z'] = (d['s'] - d['s'].mean()) / d['s'].std()
d = d.merge(rets, on=['ticker', 'day']) # rets: ticker, day, xret (%)
fit = smf.ols('xret ~ z', data=d).fit(cov_type='cluster',
                                     cov_kws={'groups': pd.factorize(d['day'])[0]})
```

C2: Critique an AI Answer (2/3) [Proposed]

Solution discussed in class

□ Conclusion of the AI assistant

- ▶ “Accuracy 74.2%, more than twice the 33% of a random guess among three classes: Qwen2.5-7B reads financial sentiment very well.”
 - ▶ “Slope +41.4 bp per standard deviation of the score, $t = +19.07$ (clustered by day): the day's headlines forecast the stock's excess return.”
 - ▶ “Qwen2.5 was never trained on these labels or return series, so the 2009–2023 backtest is fully out of sample.”
- The answer contains **three** errors typical for sentiment backtests; the code runs and the numbers are right
- ▶ (a) Find each error: name the line of code or the sentence, and explain why it is wrong
 - ▶ (b) Say how you would detect it without knowing the answer: a class count, a timestamp check, a release date, a memory test
 - ▶ (c) Write the corrected code or the corrected statement, with the numbers it produces
 - ▶ (d) Log each error as in [AI_ERRORS.md](#): what the AI answered, what was wrong, how you found out, the correct version

C3: Project Seed from Our Research: LLM-VaR on Course Data (task 1/2) [Proposed]

Solution discussed in class

Task

▣ Context and assumptions

- ▶ Lecture 15 case study [Pele et al. \(2026\)](#): an LLM forecasts next-day VaR (Value at Risk) 1% and ES 2.5% from past returns written as digit strings (encoding of [Gruver et al. \(2023\)](#)); code of the paper: [Pele \(2024\)](#)
- ▶ Published setup (Sections 3.2.2 and 4.1): rolling window w , 1024 sampled next-day returns per day, $\pi = 2$ digits, temperature $\tau = 0.7$, $\alpha_{LLM} = 0.95$, $\beta_{LLM} = 0.35$, Gaussian kernel smoothing of the draws
- ▶ Data: daily log-returns of the S&P 500, the DAX and Bitcoin (course series); model: Qwen2.5-7B [Qwen Team \(2024\)](#), open weights, in its base and instruct versions; test period: only days after the model's release
- ▶ Model: B7 (does the LLM remember the market?) for the post-release test period; VaR and ES backtests as in Seminar 8

C3: Project Seed from Our Research: LLM-VaR on Course Data (task 2/2) [Proposed]

Solution discussed in class

Task

What to do

- ▶ (a) For each asset and each test day, encode the previous $w = 30$ returns with the published rules, draw 1024 next-day returns from the base model and compute VaR 1% and ES 2.5% from the smoothed draws
- ▶ (b) Repeat (a) with $w = 150$, and repeat both windows with the instruct version of the model
- ▶ (c) Compute the benchmark forecasts of a GARCH(1,1) model with Student- t innovations ($\nu = 5$) re-estimated on a rolling 250-day window, as in the paper
- ▶ (d) Backtest every forecast series: failure rate, POF p -value [Kupiec \(1995\)](#), traffic-light $\Phi(z)$, CC p -value [Christoffersen \(1998\)](#) for VaR 1%, and Z_2 [Acerbi & Szekely \(2014\)](#) with a simulated p -value for ES 2.5% (Seminar 8)
- ▶ (e) Rerun (a) with five different random seeds and report the range of the failure rate, to separate sampling noise of the 1024 draws from model error

What to report

- ▶ A results table in the format of the lecture's Key Results slide (asset $\times$ model $\times$ window), an exceedance chart for the best LLM series, and a Quantlet folder (code, Metainfo, charts)
- ▶ Interpretation: does the short-window advantage reported for GPT-3.5 carry over to an open-weight model and to data after its release, and does the instruct version behave like the RLHF-aligned GPT-4 models of the paper?

Common Pitfalls

- ▣ Reporting accuracy without the majority-class benchmark and macro-F1
- ▣ Evaluating a model on its own training data (FinBERT on PhraseBank)
- ▣ Comparing two classifiers through separate intervals instead of a paired test
- ▣ Matching a date-only headline to the same-day return and calling it a forecast
- ▣ Backtesting an LLM on news older than the release of its weights
- ▣ Choosing the prompt, the threshold or the model after seeing the test results
- ▣ Ignoring costs and the number of clusters: four trades a day, only 16 stocks
- ▣ Treating a model score as data: slopes are attenuated, and two model scores may share their errors

Take-Home Assignment (Team Project)

- Choose a text source not covered in class: BVB (Bucharest Stock Exchange) announcements, Romanian financial news, central-bank statements or crypto posts
- Label at least 200 texts yourselves (two annotators, report the agreement) for the sentiment about the asset mentioned; tune on one part and evaluate only on a held-out part (or out-of-fold)
- Compare LM, FinBERT, an open LLM and embeddings + logistic regression: accuracy and macro-F1 with CI, McNemar tests; for Romanian texts say how you apply the English LM and FinBERT (translation or a multilingual model)
- Build a signal with correct timing; report the period after the LLM's release separately
- Declare AI use in [AI_USE.md](#) and log every AI error you catch in [AI_ERRORS.md](#) (at least three per project, as in C2)
- Deliver a Quantlet folder (code, Metainfo, charts) and three slides: numbers, interpretation, limitations

Further Study on Quantinar

Advanced courses on quantinar.com

- ▣  Natural Language Processing
- ▣  The Next Word Problem
- ▣  DELTA
- ▣  LDA Extensions

Code and data

- ▣ Seminar notebook (Colab); Quantlets: github.com/danpele/MFM/tree/main/Quantlets/Ch_15

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